Human Forms of Reasoning

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Abstract:

This paper is about kinds of reasoning that are employed in human cognition. Drawing a distinction between direct (e.g., Modus Ponens, Conjunction Elimination) and indirect (e.g., Hypothetical Reasoning, Reductio ad Absurdum) forms of reasoning, we propose that the latter may be uniquely characteristic of human cognition. We review comparative evidence indicating that non-human great apes display reasoning abilities, while noting that the extant data are consistent with the deployment of direct inferential strategies alone. Evidence for the use of indirect forms of reasoning in human cognition comes from natural language semantics/pragmatics and from research on psychological defense mechanisms. Indirect forms of reasoning require access to earlier stages of computation, i.e. access to the suppositional states in the derivational history of the reasoning in question, which, by hypothesis, relies on extended involvement of the procedural memory. We tentatively suggest that the uniqueness of humans in various forms of reasoning is related to a boost in the procedural memory within the evolution of *homo sapiens*.

Keywords: reasoning, logic, symbolic cognition, human evolution

1. Introduction

This paper is about kinds of reasoning that are employed in human cognition. Drawing a distinction between direct (e.g., Modus Ponens, Conjunction Elimination) and indirect (e.g., Hypothetical Reasoning, Reductio ad Absurdum) forms of reasoning, we propose that the latter may be uniquely characteristic of human cognition. We review comparative evidence indicating that non-human great apes display reasoning abilities, while noting that the extant data are consistent with the deployment of direct inferential strategies alone. Evidence for the use of indirect forms of reasoning in human cognition comes from natural language semantics/pragmatics and from research on psychological defense mechanisms. Indirect forms of reasoning require access to earlier stages of computation, i.e. access to to the suppositional states in the derivational history of the reasoning in question, which relies on extended involvement of the procedural memory. We tentatively suggest that the uniqueness of humans in various forms of reasoning is related to a boost in the procedural memory within the evolution of *homo sapiens*.

Explicating the distinctive nature of human reasoning requires some understanding of different types of reasoning in logic, where the central concept is that of validity. Logic is about distinguishing those arguments that are valid from those that are not.

(1) Premise 1. If John is at home, his lights are on.

Premise 2. John is at home.

Conclusion: John's lights are on.

There is some kind of indefeasibility about the argument in (1). Once the premises are accepted as true, one is effectively compelled to accept the truth of the conclusion. What makes this argument valid? It is nothing personal about John, or about him being at home. Neither is it about the lights being on at the relevant moment. Here, validity is a form-related concept determined exclusively by the logical vocabulary in the premises. The argument in (1) is valid because it instantiates the Modus Ponens schema (also known as *Conditional Elimination*, or "→E") shown in (2), where "φ" and "ψ" are metavariables over simple and complex sentences.

(2) Modus Ponens

$$\frac{\varphi \rightarrow \psi \qquad \varphi}{\psi} \rightarrow E$$

Modus Ponens is a local rule in the sense that its application requires consideration only of the two immediately preceding propositions. Forms of inference of this kind are typically classified as *direct* reasoning, insofar as the conclusion follows immediately from the most recent premises. However, logic also permits other forms of reasoning. Suppose, for instance, that I wish to determine whether John's cat is happy. I know that if John's lights are on, then his cat is happy, and I also know that if John is at home, then his lights are on. From these premises, I may conclude that John's cat is happy, *provided that John is at home*. In this style of argument, premises are evaluated within the scope of a temporary supposition, which is subsequently discharged at the level of the conclusion.

(3) Supposition: John is at home

Premise 1. If John is at home, then his lights are on.

First Interim Conclusion. John's lights are on.

Premise 2. If John's lights are on, then his cat is happy.

Second Interim Conclusion: John's cat is happy.

Conclusion: If John is at home, then his cat is happy.

The form of this reasoning in (3) can be represented in two steps. There are two rules of Modus Ponens that must be applied before we can derive the overall conclusion. The suppositional status of the sentence "John is at home" is indicated with the superscript "1".

(4)

$$\dfrac{\dfrac{p^{1} \qquad p \rightarrow q}{q} \qquad q \rightarrow r}{r}$$

We now eliminate the suppositional status of *p* by inserting it as the antecedent of a conditional at the lowest level in the derivation. In this form of reasoning, we conditionalize an interim conclusion to a supposition, which is why such a reasoning is sometimes called *Hypothetical Reasoning*. The symbol "$\rightarrow I^1$" should be read as the rule of conditional introduction targeting the premise indexed with 1. The elimination of a suppositional premise is indicated with a strikethrough over the premise.

(5)

$$\dfrac{\dfrac{\dfrac{\cancel{p^{1}} \qquad p \rightarrow q}{q} \qquad q \rightarrow r}{r}}{p \rightarrow r}\ \rightarrow I^{1}$$

The general syntactic schema of the rule of Conditional Introduction is given in (6). Observe that, unlike Modus Ponens, this rule requires access to earlier stages of the argument in order to identify the supposition that is to be eliminated. In other words, the conclusion is *not* a function of the last steps in the derivation. This is why this form of reasoning is called *indirect*: Conditional Introduction is an indirect form of reasoning. In this proof, the sentence φ (indexed with j to express its suppositional status) is said to be *discharged* by the application of the rule of Conditional Introduction targeting it.

(6) Conditional Introduction

$$\dfrac{\begin{matrix}\cancel{\varphi}^{j}\\ \vdots\\ \psi\end{matrix}}{\varphi \to \psi}\,\to\mathrm{I}^{j}$$

The kind of rules illustrated in (2) and (6) are called Gentzen Rules, named after Gerhard Gentzen, who introduced them (Gentzen, 1934; see also Braine, 1978; Rips, 1994; Braine & O'Brien, 1998 for syntactic approaches to human reasoning based on Gentzen's system). Conditional Introduction is not the only form of indirect reasoning in this system. Suppose you suspect that the set of real numbers is countable. This would mean that there exists a bijective function from the set of natural numbers, which is countable, to the set of real numbers. By applying the diagonalization argument devised by Georg Cantor, however, you come to see that no such bijection can exist. The assumption that the set of real numbers is countable leads to a contradiction, from which you correctly conclude that the set of real numbers is not countable. (This is a type of reasoning known as *Reductio Ad Absurdum*). Underlying the reasoning in this example is the rule of Negation Introduction targeting an assumption (“the set of Reals is countable”) made at an earlier stage in the derivation, which is an indirect form of reasoning. This rule has the following schema, where we use “$\perp$” to express contradiction.

(7) Negation Introduction

$$\frac{\begin{array}{c}\cancel{\varphi}^{j}\\ \vdots \\ \bot\end{array}}{\neg\varphi}\;\neg\mathrm{I}^{j}$$

We can derive the rule of Negation Elimination by postulating an entailment from "¬¬φ" to "φ". What we do is choose "$(\neg\psi)^{j}$" as the input for the rule of Negation Introduction, as a result of which we derive "¬¬ψ", which entails "ψ" by our assumption. This is the rule of Negation Elimination. In other words, Negation Elimination runs Negation Introduction as a subroutine, which is why we consider it to be an indirect form of reasoning.[1]

Let us also note that Conditional Elimination is not the only direct inference rule in logic. We can obtain a conjunction by combining conjuncts and we also can eliminate conjuncts from a conjunction. These are the rules of Conjunction Introduction and Conjunction Elimination respectively. Note that Conjunction Elimination can target any of the conjuncts.

(8) a. Conjunction Introduction b. Conjunction Elimination

$$\frac{\varphi \qquad\qquad \psi}{\varphi \wedge \psi}\;\wedge\mathrm{I} \qquad\qquad\qquad \frac{\varphi \wedge \psi}{\{\varphi, \psi\}}\;\wedge\mathrm{E}$$

We now have logical rules that introduce or eliminate operators and connectives. Some of these rules are direct, in that they depend only on the results of the most recent steps in a derivation. Others are indirect, in that they target assumptions introduced at earlier stages of the derivation. We can summarize these observations as follows:[2]

[1] As a matter of fact, there is also a direct way of expressing the rule of Negation Elimination. Given some formula of the form "¬φ", we match it with a formula of the form "φ" to obtain a contradiction and, by principle of explosion, from contradiction, any formula, including "φ", follows, which is how we end up eliminating the negation operator. Paraconsistent systems of logic have been developed to avoid the principle of explosion as a legitimate property of logic (Priest, 2002). In fact, as we will see at the end of Section 3, human mind is very much disturbed by contradictory states, which leads to cognitive dissonance (Festinger, 1957). We find the indirect way in which negation is eliminated in the main text more natural and easier to work with, although it must be said that nothing crucial hinges on this assumption. We will not make use of the rule of Negation Elimination in the rest of the paper.

[2] We do not include rules for introducing and eliminating disjunction in our list due to their unnaturalness. Given some formula "φ", we can derive "φ ∨ ψ" by a weakening rule of logic. It is, however, difficult to see how such a

Direct Forms of Reasoning:

Conjunction Introduction, Conjunction Elimination, Conditional Elimination

Indirect Forms of Reasoning:

Conditional Introduction, Negation Introduction, Negation Elimination

The central claim of this paper is that *only humans mentally compute indirect forms of reasoning*, the type of reasoning that requires access to earlier stages of the computation. We suggest, tentatively, that indirect reasoning requires extended involvement of the procedural memory, i.e. a boosted procedural memory, which is possibly a uniquely *homo sapiens* property (see Section 4 for a detailed discussion of this point).

After noting the counterintuitive character of Hilbert proof systems, Rips (1994, p. 38) observes that "[p]eople often assume certain sentences are true 'for the sake of argument' in order to simplify their thinking," a feature expressed in the Gentzen system by indirect rules of reasoning. Experimental evidence that Gentzen rules provide a good approximation of the rules of internalized logic comes, in part, from the distinction the system draws between the representation of Modus Ponens (MP) and Modus Tollens (MT): $\{p \rightarrow \neg q, p\} \vdash \neg q$ (MP) and $\{p \rightarrow \neg q, q\} \vdash \neg p$ (MT). While MP is a direct application of the rule of Conditional Elimination, the derivation of MT is more involved.

rule would be of any use to an animal. In order to eliminate disjunction in a formula of the form "$\varphi \vee \psi$", we turn each disjunction into a suppositional state (e.g., φ^i and ψ^i) and derive a formulate χ from each of them so that we can replace "$\varphi \vee \psi$" with "χ", thereby eliminating disjunction. However, entertaining double suppositional states does seem to be cognitively costly. Moreover, typically, we cannot distinguish between a disjunctive sentence of the form "$\varphi \vee \psi$" from a conjunction of possibilities, i.e. Poss(φ) $\wedge$ Poss(ψ), in an experimental setting. For instance, Engelmann et al. (2021, p. 1377) provide experimental evidence that "chimpanzee thought is not limited to what is, but also involves reasoning about what could be the case." In their experiments, two containers were placed on the top of two platforms where a rope, which extended into the testing room, was treaded through a loop on each platform. When the chimpanzees knew where the food was, either because the containers were transparent or because they observed the experimenter put the food into the opaque container, they were more likely to pull the only end of the string that provided immediate access to the platform with the baited container. However, when the chimpanzees did not know where the food was, they were more likely to pull the string on both ends to gain access to both of the platforms at the same time. Engelmann et al. (2021) suggests that the reasoning underlying the behavior of pulling the sting on both ends is a disjunctive one: has(container1, food) $\vee$ has(container2, food). However, the reasoning underlying this behavior can also be characterized as a conjunction of possibilities: Poss(has(container1, food)) $\wedge$ Poss(has(container2, food)), where "Poss(φ)" entails something that the state of affairs described by "φ" cannot be ignored. Distinguishing between these two possible interpretations of the experimental results on ape cognition is no trivial task. In fact, Rescorla (2009) argues that the effects of disjunctive syllogism can be captured via a system of probabilistic reasoning over cognitive maps.

(9)
$$\dfrac{\dfrac{\dfrac{\cancel{p}^{1} \qquad p \rightarrow \neg q}{\neg q} \qquad q}{\bot}}{\neg p}\ \neg I^{1}$$

Experimental studies comparing Modus Ponens (MP) and Modus Tollens (MT) have consistently shown that participants process MP both more accurately and more rapidly than MT (Evans et al., 1993; Girotto et al., 1997; Li et al., 2014). This asymmetry is consistent with their status in Gentzen-style deduction, lending support to the view that such systems capture important aspects of human reasoning.[3] Moreover, there is evidence from research in natural language semantics that human grammar incorporates a natural logic as a subcomponent, which has been argued to account for complex patterns of acceptability judgments associated with natural language expressions (von Fintel, 1993; Gajewski, 2002; Chierchia, 2013).

Before we discuss direct forms of reasoning in primate cognition, it is important to distinguish between simple and complex schemas of inference. The rules we have given above are basic forms of reasoning in that they involve *addition and elimination* of connectives. It is also possible to summarize complex interactions between these basic rules as valid schemas. For instance, the kind of proof that we have seen in (5) can be summarized as in (10), which is called Transitive Inference.

[3] This is not to say that the Gentzen system is the only account of human reasoning capable of explaining the cognitive distinction between Modus Ponens (MP) and Modus Tollens (MT). An important alternative is the Mental Models Theory, which offers a semantic characterization of reasoning. According to this theory, reasoners construct mental models of the premises and evaluate validity by searching for alternative models that constitute counterexamples to the conclusion (Johnson-Laird, 1983; Johnson-Laird & Byrne, 1991). For example, Girotto et al. (1997) argue that the difference between MP and MT arises because MT requires the representation of a negated proposition in the initial model of the conditional, which is cognitively costly, and they present experimental evidence in support of this claim.

Although the debate between Mental Models Theory and Mental Logic has often been framed as a competition between mutually exclusive accounts, the two approaches need not be viewed as incompatible. It is plausible that mental models provide the interface through which the inferential machinery posited by Mental Logic interacts with other cognitive systems, such as imagination, planning, and decision-making. On this view, the two theories capture different aspects of human reasoning and should be regarded as complementary rather than competing accounts.

(10) Transitive Inference

$$\frac{\varphi \rightarrow \psi \qquad \psi \rightarrow \chi}{\varphi \rightarrow \chi}$$

Looking at the schema in (10), we may come to think that it instantiates a direct form of reasoning given that the consequence appears to be a direct consequence of the last two premises. This impression is misleading, however. It is important to note that what we call Transitive Inference is obtained by the application of the rule of Conditional Introduction subsequent to the two instances of Conditional Elimination, as was shown in (5). In other words, the representation of Transitive Inference in (10) is not indicative of its internal complexity. Therefore, as far as this paper is concerned, Transitive Inference is a complex form of indirect reasoning since it involves direct and indirect forms of reasoning as its constituents. In a sense, Transitive Inference is a summary of some complex interactions between basic forms of direct and indirect reasoning. This is an important consideration to keep in mind in evaluating the internal complexity of an argument schema.

2. Direct Forms of Reasoning in Ape Cognition

In this section, starting with Conjunction Introduction, we provide evidence that apes can mentally implement direct forms of reasoning. We must, first, distinguish between conjunctive thoughts and conjunctive reasoning. Conjunctive thoughts may be widespread in ape cognition—and, more generally, in animal cognition—serving to represent external objects, events, or situations. Hurford (2007, p. 114), for instance, suggests that "part of what a baboon perceives when it contemplates a scene containing a crouching lion and a rock" can be represented as "$\text{LION}_{baboon}(x)$ & $\text{CROUCH}_{baboon}(x)$ & $\text{ROCK}_{baboon}(y)$" where the index "baboon" is intended to indicate that these concepts are relativized to the baboon cognition, i.e. whatever the baboon cognizes when she cognizes some animal as a lion or as crouching, and the variables x and y stand for "the mental indices of these objects in the baboon's brain as it tracks them (probably realized as activity in its posterior parietal cortex)". An animal that can track objects and register their properties can be attributed "a (limited) capacity for logical conjunction", limited presumably by the constraints on the number of objects it can track (see the discussion in Hurford, 2007, pp. 90 – 96). Similarly, Tomasello (2014) suggests that a chimpanzee approaching a banana tree with the intention of foraging attends to a set of situations that are relevant to her decision making. These situations may include: (a) that there are many bananas are in the tree, (b) that the bananas are ripe, (c) that the bananas are reachable by climbing, (d) that no predators are nearby etc ... and it is a conjunction of these situations that play a role in the execution of foraging behavior of the chimpanzee. These sub-situations that are relevant to implementing a specific behavior are "all present in a single visual image." (p. 11).

Conjunctive reasoning goes beyond logical (here, conjunctive) analysis of the stimulus. Conjunctive reasoning should involve integrating distinct types of information from the environment that are causally relevant to the behavior. Experimental evidence for conjunctive reasoning in great apes can be found in the future-oriented tool manipulation of a zoo-housed Sumatran orangutan by the name of Riau (Mulcahy, 2018). In an enrichment task involving the use of a long and heavy tool to rake-in the rewards that were out of his reach, Riau was observed to spontaneously hang up the tool into gaps of the mesh fence surrounding his enclosure, presumably to avoid the cost of re-carrying the tool from the ground when it would later be needed. In these experiments, twelve pieces of fruit (apples and oranges) were placed in an open container outside Riau's enclosure, where they remained clearly visible to him throughout the task. During each trial, the experimenter selected two fruits from the container and placed them inside a metal ring positioned on a wooden platform fixed to the exterior of the enclosure fence. To obtain the fruit, Riau had to retrieve the ring's contents by pulling it toward himself with a stick. Mulcahy (2018) ran two experiments on Riau's tool-oriented behavior, one to test the hypothesis that this behavior is contingent on the *future* availability of the reward and a second experiment to show that the size and/or shape of the tool is a decisive factor in deciding to implement this behavior. The first experiment showed that, in 59 of 60 trials, Riau wedged the tool into the narrow gaps of the fence when food remained in the container for later consumption—that is, *when he could see that the tool would be needed later* to retrieve additional food items. Moreover, he almost never exhibited this tool-oriented behavior *when he could see no future use for the tool* (1/12). The second experiment showed that when given a smaller and lighter tool, he never exhibited this securing behavior.[4] The conjunctive reasoning underlying this

[4] The second experiment also involved a control where Riau manipulated the large tool. In all cases where there was future use for the tool, Riau wedged it inside the gaps of the fence (30/30). In those cases where there was no more reward to be retrieved, he did not exhibit this securing behavior (0/6). This strenghtens the conclusions of the first experiment.

behavior can be represented as in (11), where *FUT* is a temporal operator that locates the event or situation denoted by the sentence at a time later than *now*:

(11) BEHAVIOR: hangs up the tool

REASONING UNDERLYING THE BEHAVIOR:

$$\frac{\text{large(stick)} \qquad\qquad \text{FUT(available(food))}}{\text{large(stick)} \wedge \text{FUT(available(food))}} \wedge\text{I}$$

To the extent that the analysis of Riau's behavior in (11) is on the right track, it provides independent evidence for sentential representations in animal cognition (see also Hurford, 2007). This is of significance in that some authors deny non-human animals the capacity to represent sentence-like structures, which means that they also deny them the capacity for logical reasoning, the kind of reasoning that is based on sentential operators. Bermúdez (2006, p. 129), for instance, claims that "an account of animal reasoning" requires that we "identify forms of reasoning at the non-linguistic level and then explain them without assuming that the animal is deploying elementary logical concepts or exploiting the internal structure of a thought", where logical concepts include conjunction, disjunction, negation, quantification etc. Given (11), however, we take it that a great ape is capable of mentally representing tensed thoughts. The tense operator is understood to locate an event or situation—namely, the denotation of a sentence—relative to the contextual variable *now*. That is, it is an operator that takes sentences as its input; hence the formalization "FUT(φ)" for a given sentence φ. In other words, tensed thoughts involve tense operators (e.g., past, future), which act on sentences as input.[5] This means that an animal that can mentally manipulate these operators can be said to mentally represent sentences as units of symbolic manipulation. There is, then, no conceptual reason why such an animal should not do logical reasoning.

[5] This is independent of whether tense is analyzed as a quantificational operator, in the spirit of Prior (1967), where "FUT(φ)" is interpreted as "∃t. φ(t) ∧ follows(t, now)", or as a referential operator, as in Reichenbach (1947) and Partee (1973), where "FUT(φ)" is interpreted as "φ(t) ∧ follows(t, now)", with the assignment function determining directly the value of the temporal variable.

Some authors claim that non-human animals exhibit no temporal cognition. Hoerl and McCormack (2019), for instance, argue that temporality in other animals is restricted to a temporal updating system that can represent only the current state of affairs. This system "operates with a model of the world that concerns the world only ever as it is at present. Other times, and how things are at other times, are not represented in the system" (Hoerl & McCormack 2019, p.2) Kaufmann and Vieira (2025), on the other hand, suggest that ape temporal life extends beyond mere updating. Drawing on evidence from the hunting and foraging behavior of Taï chimpanzees, they argue that great apes systematically and flexibly integrate temporal information in the service of goal-directed behavior, which they take to constitute evidence for temporal cognition. For instance, to obtain figs, a highly sought-after fruit in the competitive environment of the Taï Forest (Côte d'Ivoire), Taï chimpanzees must arrive at feeding sites early in the morning. Building on Janmaat (2019), Kaufmann and Vieira (2025) note that "… regardless of how far away the fig trees were from the nesting site, the foraging chimpanzees would arrive at more or less the same time in the morning for breakfast. They would do this by adjusting their time of departure and their travel speed given the distance and the type of journey separating their nesting site from the figs," the distance between which can be as much as 900 meters. Here, departure time and travel speed appear to function as variables that chimpanzees can flexibly adjust in the pursuit of a temporally defined goal. Evidence for temporal thought in great apes comes from *prospection*, which has been defined as *thinking about the future* (Suddendorf et al., 2018). In Riau's tool-hanging, for instance, the action makes sense only from the perspective of a future state in which the food items in the container are brought into the metal ring. It is, then, reasonable to suppose that Riau entertains a thought of the form "it will be the case that food is available," which we abbreviate as "FUT(available(food))". Such a thought enables the anticipatory behavior of hanging up the tool at time *t*, even though the presence of a hanging tool is of no use to Riau at *t* and becomes useful only at a later point. Moreover, the fact that this behavior is not exhibited when there are no food items in the container suggests that it cannot be reduced to a mere habit. Evidence of similar anticipatory behavior can be found in the case of Santino, a chimpanzee housed at the Furuvik Zoo in Sweden, who has been observed to cache stones and pieces of concrete early in the morning for use as weapons against visitors later in the day (Osvath, 2009, Osvath & Karvonen, 2012). In our view, such anticipatory behavior, facilitated by prospection, provides evidence for tensed cognition in great apes. In fact, drawing on such evidence, Bayırlı (2026) identifies a set of grammar rules underlying temporal sentences in ape cognition.

Having discussed conjunctive reasoning in great apes, we now turn to the question of whether they are capable of conditional inferences in accordance with Modus Ponens, which is a direct form of reasoning. Tomasello (2016, p. 62) finds evidence for a version of this type of reasoning in the tool use of great apes, observing that they "make inferences all day every day of the Modus ponens type: Poking with a stick causes ants emerge. I poke. This causes ants to emerge." Importantly, there may be an alternative way of thinking about the tool use in question. In this case, tool use may instantiate a type of *kind reasoning*: A great ape coming across a stick may entertain the thought that this is the *kind of thing* that makes ants available. In this sense, kind reasoning relates objects (the stick) to situations (the availability of ants). This reasoning should be distinguished from Modus Ponens: The conditional statement inside Modus Ponens relates situations to situations, where situations are construed as the denotation of sentences. Better evidence for Modus Ponens in ape thought would be a case in which an object leads to a situation only if some other situation also holds, a type of reasoning that we shall call *conditionalized kind reasoning*. The details of tool use by chimpanzees in Goualougo, the Republic of Congo, provide some evidence for conditionalized kind reasoning in great apes.

Goualougo chimpanzees make and use tools for fishing termites located (invisibly) inside an earth mound (see Byrne et al., 2013 for a detailed description). The process begins with *nest puncturing,* where the chimpanzee pushes a sturdy stick into the ground to open a tunnel that enables access to the termite colony.[6] This is followed by *termite fishing* during which the chimpanzee straightens the brush fibers of the fishing probe each time before inserting it into the nest. The next step is *tunnel clearing*, in which the chimpanzee "reverses the orientation of a brush-tip tool and uses the blunt end to clear loosened soil from the entrance of the fishing hole." (Byrne et al., 2013, p. 56). What is important for our purposes is that the subroutine involving termite fishing *and* tunnel clearing is repeated several times in this order. In fact, Byrne et al. (2013) provide a minute-by-minute description of the loop on this subroutine. After nest

[6] The end point of the tool is smelled or visually inspected to see if access to the colony has been achieved. The execution of such test routines provides evidence that what the chimpanzee is doing goes beyond a linear associative behavioral sequence learnt from past contingencies. The chimpanzee represents a goal state and makes changes to the world to attain it (see Byrne et al. 2013 for further discussion).

puncturing at 9:07 and termite fishing at 9:15, the following steps are taken by the chimpanzee: (9:27 tunnel clearing, 9:28 termite fishing), (9:28 tunnel clearing, 9:28 termite fishing), (9:32 tunnel clearing, 9:33 termite fishing)… The loop on this subroutine suggests that the chimpanzee is aware that the stick provides access to ants only if the brush fibers are straight and the tunnel is clear. This reasoning can be represented as in (12)

(12) REASONING UNDERLYING TERMITE-FISHING BEHAVIOR

$$\frac{\begin{array}{ll}\text{straight(fiber)} \wedge \text{clear(tunnel)} \rightarrow \text{available(ant)} & \text{(by the affordances of the stick)} \\ \text{straight(fiber)} \wedge \text{clear(tunnel)} & \end{array}}{\text{available(ant)}} \rightarrow E$$

The reasoning underlying this complex behavior is an example of the conditional kind reasoning that we have discussed above. Note that the chimpanzee behaves selectively to bring about the state of affairs denoted by each of the conjuncts inside the conjunctive antecedent "straight(fibers) ∧ clear(tunnel)[7]". The loop on the subroutine involving termite fishing and tunnel clearing is the chimpanzee trying to sustain this conjunctive state. This provides evidence that the conjunctive analysis of the conditional clause we have adopted in (12) reflects the pieces of reasoning underlying complex termite-fishing behavior more or less accurately.

It should be noted that there are important differences between the conditional statement in (12) and the material implication of formal logic. The states expressed by the two clauses within the conditional are linked by a relation of *affordance*. In formal logic, however, no such connection is required between the sentences joined by material implication. This is unsurprising: the material analysis of conditionals in natural language is well known to yield unintuitive results, often referred to as the paradoxes of material implication. For instance, for any two propositions *p* and *q*, either "p → q" holds or "q → p" holds, which does not feel intuitive at all. Similarly, if one wishes to negate the claim that "if Santa Claus lives in Reykjavík, then he has no beard," one must accept that Santa Claus is a bearded individual living in Reykjavík, given the logical

[7] It is not suggested here that the chimpanzee in question has the human concepts of *straight*, *fiber, tunnel* etc. Imitating Hurford's notation, we could index each of these concepts with the word *chimpanzee* to highlight the fact that these concepts should be relativized to chimpanzee cognition, e.g., $\text{CLEAR}_{\text{chimpanzee}}(\text{tunnel}_{\text{chimpanzee}})$.

equivalence between “$\neg(p \rightarrow q)$” and “$p \wedge \neg q$” (see Egré & Rott, 2021 for a lengthy discussion of these issues). Explaining the *necessary connection* between the antecedent and the consequent is the main aim of modal logics that are intended to characterize conditionals in natural languages (Stalnaker, 1968; Kratzer, 1986 among many others). For instance, a sentence like “if an object is unsupported then it falls” is understood to mean something like “by the laws of physics, if an object is unsupported then it falls”, or more precisely as “given a possible situation s in which the laws of physics are operative and an object is unsupported, the object falls in s”. In our examples, this connection between the antecedent and the consequent is formed via the affordances of the stick used by the chimpanzee.[8]

It has been suggested that many animals are capable of transitive inference (see Lazareva, 2012 and Gazes & Lazareva, 2021 for overview and see Vasconcelos, 2008 for a defense of explanations based on associative learning procedures). In a typical experimental procedure designed to test for such inferences, an animal is trained on four pairs drawn from a set of five stimuli, each associated with differential reward contingencies: (A+B–), (B+C–), (C+D–), (D+E–), where “+” means the associated stimulus is rewarded and “–” indicates the absence of any reward.[9] Once a predetermined criterion of correct performance is achieved, animals are tested on the pair (B, D), with which they have had no prior experience. Even though B and D are rewarded with equal frequency in the training, many animals tend to choose B over D. This result has sometimes been analyzed as an instance of transitive inference, where given the prominence of B over C and the prominence of C over D, the animal under study infers the prominence of B over D and behaves accordingly.

Do such experiments genuinely capture instances of logical transitive reasoning in animals? As noted at the end of the previous section, transitive inference, in its logical instantiation, can be viewed as a schematic shorthand for two applications of Modus Ponens followed by Conditional Introduction. As we have seen, the rule of Conditional Introduction begins with a supposition that we typically do not know to be true (Hypothetical Reasoning) or

[8] We will not comment on the rule of Conjunction Elimination in ape cognition, which is a weakening rule in that the conjunction is always logically stronger than its conjuncts. If an ape represents a stimulus conjunctively, as Hurford and Tomasello claim they do, then Conjunction Elimination may be operative in the capacity to attend to non-salient properties of a stimulus. Given a stimulus which is analyzed as “$p \wedge q$”, where p is salient and q is not, the ape can derive q and behave in a way that is solely sensitive to q being the case.

[9] More than three pairs are needed in the training phase so as to guarantee that the pair that is tested does not contain a stimulus that has always been rewarded or one that has never been rewarded.

may even know to be false (Counterfactual Reasoning). We then derive the consequences of this supposition, only to discharge it at the end of the derivation as the antecedent of a conditional statement. However, what is termed *transitive inference* in the experiments described above is better understood as the placement of stimuli along an ordered continuum and has little to do with the evaluation of suppositional states of affairs. The relevant representations concern "orderly relations among stimulus items" (Dusek & Eichenbaum, 1997, p. 7133). In other words, they are representations of objects arranged along a line, that is, linear representations. In this respect, such inferences appear more geometric than logical. We therefore maintain that these experiments do not provide convincing evidence that non-human animals are capable of indirect forms of reasoning in the logical sense.

The central idea of this paper is that indirect forms of reasoning are unique to human cognition. This conjecture does not entail that non-human animals are capable of sentential representations and/or direct forms of reasoning. It is conceivable that both direct and indirect forms of reasoning emerged in the evolutionary history of Homo sapiens and are not shared with any other ape species or, indeed, with any other animals. However, it is evolutionarily more plausible that the infrastructure for reasoning, in the form of basic reasoning capacities, was already present in ape cognition before the emergence of the genus Homo. On this view, human cognition did not invent reasoning in all its complexity; rather, it developed novel ways of exploiting pre-existing reasoning capacities. For this reason, we consider it important to provide evidence for sentential representations and basic forms of reasoning in ape cognition. Whether non-human great apes possess sentential representations used by mental operations remains an understudied question that deserves further investigation.[10] In the next section, we present evidence that humans, and plausibly humans alone, are capable of indirect reasoning.

[10] One reason for the lack of interest in sentential representations in ape cognition may be that such representations are not used communicatively by other apes in the way they are used by humans through public languages. It is important, however, to distinguish between possessing a mental representation and expressing the content of that representation in a public language in order to influence the beliefs of other members of a community by expressing informative intentions. For instance, Scott-Phillips and Heintz (2023) argue that human communication goes beyond merely expressing intentions. It is *Gricean* in that by uttering a sentence *S* with content *p* to an audience *A*, a speaker intends that *A* recognize the speaker's intention that *A* come to believe that *p*. According to Scott-Phillips and Heintz (2023), the capacity to form such higher-order informative intentions lies at the heart of human communication and is unique to humans.

3. Indirect Forms of Reasoning in Human Cognition

In this section, we present evidence for the implicit use of indirect forms of reasoning in human cognition. We identify such reasoning in conditional representations, pretense play as well as in the psychological defense mechanism of Rationalization.

The rule of Conditional Introduction involves reasoning under a supposition. A speaker's attitude towards this supposition may vary: the speaker may consider it to be a live option about the world (*hypothetical reasoning*) or the speaker may consider it to be unlikely to be the case (*counterfactual reasoning*). This distinction has grammatical ramifications reflected in the categories of indicative vs subjunctive conditionals. Consider (14a) and (14b) from Adams (1970, see also the discussion in von Fintel, 2011)

(13) a. If Oswald didn't kill Kennedy then someone else did. (Indicative)

b. If Oswald hasn't killed Kennedy then someone else would have. (Subjunctive)

Given that Kennedy was assassinated, (13a) is readily judged to be true. By contrast, accepting (13b) as true requires additional, non-trivial assumptions about American political life. Karttunen and Peters (1979) characterize the distinction between indicative and subjunctive conditionals in terms of possibility relative to what is known. In the indicative mood, a sentence of the form "If A, then B" highlights the fact that *A* is epistemically possible. In the subjunctive mood, the same sentence puts focus on the fact that it is epistemically possible that *A* is false. In any event, by hypothesis, the ability to represent hypothetical and counterfactual scenarios and reason within such suppositional states is a capacity that enables humans to distance themselves from the actual states of affairs, a capacity enabled by the rule of Conditional Introduction. This is, indeed, a powerful mode of thinking. In response to various complaints about the reliability of suppositional reasoning, Steiner (1998, p. 226), in *After Babel*, writes: "Hypotheticals, 'imaginaries', conditionals, the syntax of counterfactuality and contingency may well be the generative centers of human speech."

It is likely that the rule of Conditional Introduction has something to do with symbolic forms of *pretense* observed in children (see Wiesberg, 2015 for an overview). Symbolic pretense should be distinguished from behavioral imitation (see Adair & Carruthers, 2022 on the role of imitation in pretend play) in that the former involves what Nicols & Stich (2000, p. 119) call *inferential elaboration*: "Inference often plays a crucial role in filling out the details of what is happening in pretense." Nicols & Stich (2000) suggest that symbolic pretense involves an initial premise, a suppositional state, within which the pretender "draws inferences about what is going on in the pretense." In other words, pretense happens in a mental workspace, what Nicols & Stich call a *possible world box*, opened up by a suppositional state and the types of reasoning rules that apply inside the box are the same as those that operate on the actual beliefs. They discuss a three-year-old child, reported in Gould (1972), who, when climbing on a jungle gym, said: "I am a pussycat. Meow, meow". After coming down from the jungle gym, the boy laid on the ground and said: "I'm dead. I am a dead pussycat… I got shooted." Here the boy being a cat is the initial premise and him being dead after getting shot within this suppositional state involves types of reasoning that apply inside and outside of the "possible world box". In (14), we provide a formal approximation of the mental processes underlying the child's behavior.[11] Note that the unique cat is dead within the suppositional state in which the child is a pussycat. It follows, then, that, within this state, the child is pretend-dead.

(14) The premise: [pussycat(the_self)][1]

$$\dfrac{\text{get-shot(the\_pussycat)} \qquad \dfrac{\forall x.\ \text{get-shot}(x) \to \text{dead}(x)\ [\text{World Knowledge}]}{\text{get-shot(the\_pussycat)} \to \text{dead(the\_pussycat)}}\ \forall\text{-E}}{\text{dead(the\_pussycat)}}\ {\to}\text{E}$$

Are there convincing examples of symbolic pretense, the kind of pretense that involves inferential elaboration, in great apes? After some thirty years of natural observation at Bossou-Nimba (Guine), Matsuzawa (2020) reports of three instances of pretense in chimpanzee wild life. In one case, a young juvenile, by the name of Flanle, picked up a grass cushion used and left by a human and started walking bipedally while wearing it. This appears to be a case of behavioral imitation, where the chimpanzee "performed deferred imitation of a human model using the

[11] We remain agnostic as to whether symbolic pretense relies on the module for *theory of mind*, the capacity for mental state attribution (Leslie, 1987, 1994; Leslie & Roth, 1993). For arguments against mental state attribution in pretense, see Lillard (1993) and Currie (1998).

cushion to carry a heavy object on the head" (Matsuzawa, 2020, p. 546). In another episode, an 8.5-year-old female, Vuavua, carried a male hyrax killed by one of the other chimpanzees into her nest, where they stayed together for one night, and groomed him with her fingers and mouth, the way infant chimpanzees are typically groomed. The third episode of pretense reported in Matsuzawa (2020) features an 8-year-old female, Ja, who was observed to carry a log while Jire, her mother, was carrying her sick sister, then aged two and a half years. The log Ja was carrying on her shoulder was quite similar in size and shape to the sick infant that her mother was carrying. Gómez (2008) notes that such mothering-like behavior towards animate and inanimate object is prevalent in primate life. Young gorillas show such mothering-like behavior towards objects even in the absence of any adult model (see, Gómez & Martín-Andrade, 2002), which is reminiscent of clumsy caring-behavior associated with new gorilla mothers. We concur with Gómez's (2008, p. 588) remark that "those rare instances of play mothering with objects observed in the wild need not in fact involve any pretend element." Rather, they are indicative of "'instinctive' patterns whose function is to trigger actual maternal behavior with real babies." In other words, such reports do not seem to require that we attribute apes the capacity for reasoning within a suppositional state of the kind we have seen in (14). We propose, as a hypothesis, that pretense, when it involves inferential elaboration, is unique to humans.

We have discussed the role that Conditional Introduction plays in various aspects of human cognition. What, then, of Negation Introduction? There is evidence that humans apply this rule unconsciously in the elimination of "nonfitting relations among cognitions" (Festinger, 1957, p. 3), where cognition, in this context, includes "any knowledge, opinion or belief about the environment, about oneself, or about one's behavior" (p. 3). The presence of contradictory cognitions—where one thought entails the negation of another that an individual simultaneously entertains—gives rise to cognitive dissonance, which motivates individuals "to try to reduce the dissonance and achieve consonance" (Festinger 1957, p. 3; see also Cooper 2005; Quilty-Dunn 2022). In one experiment (Festinger & Carlsmith 1959), participants were asked to express a positive evaluation (e.g., that a task was interesting, intriguing, or exciting) to others about a boring and monotonous activity they had completed, a procedure known as forced compliance. When later asked for their honest opinions, participants who were paid one dollar expressed more positive attitudes toward the task than those who were paid twenty dollars. Festinger and Carlsmith suggest that the group receiving minimal compensation reduced dissonance by revising their belief about the task. How exactly? Quilty-Dunn (2022) suggests that the generation of dissonance is a result of the following steps of reasoning: (1) The task is not worthwhile, (2) If the task is not worthwhile, then I am incompetent for freely choosing to engage in it and (3) I am incompetent for freely choosing to engage in this task. Below, we represent this dissonance-inducing reasoning in Gentzen form ("s" denotes the self and "t" the task).

(15)

$$\dfrac{\dfrac{\dfrac{\text{competent}(s) \rightarrow [\text{willingly.do}(s, t) \rightarrow \text{worthwhile}(t)] \qquad \text{competent}(s) \quad [\text{by EGO}]}{\text{willingly.do}(s, t) \rightarrow \text{worthwhile}(t)}{\rightarrow}E \qquad \text{willingly.do}(s, t)\ [\text{World Knowledge}]}{\text{worthwhile}(t)}{\rightarrow}E \qquad \neg\,(\text{worthwhile}(t))\ [\text{World Knowledge}]}{\perp}$$

The participant has reached an *impasse*, a contradictory mental state, which is the source of cognitive dissonance. What must be done inside such a state is find a way to eliminate this contradictory state. This can be achieved by turning one of the sentences in the reasoning into a supposition so that the contradictory thought can be relativized to this suppositional state. In this way, the contradictory state need not be the final state of the reasoning and, more importantly, the newly obtained suppositional sentence can be "blamed" for the rise of contradiction. For instance, the participant may give up on the idea of competence. That is not going to work, however. *Competence* is one of those properties of our "self-concept" that we need to protect (Aronson, 1992), together with *Stability* and *Morality.* Instead the participant unconsciously turns the belief that "the task is not worthwhile" into a supposition. Moreover, since this supposition has led to a contradictory state, it must be false: It is *not* the case that the task is not worthwhile, from which it follows that the task is worthwhile (as "¬¬φ" entails "φ").

(16) $$\dfrac{\dfrac{\dfrac{\dfrac{\text{competent(s)} \rightarrow [\text{willingly.do(s, t)} \rightarrow \text{worthwhile(t)}] \qquad \text{competent(s)} \quad [\text{by EGO}]}{\text{willingly.do(s, t)} \rightarrow \text{worthwhile(t)} \qquad \text{willingly.do(s, t)}\ [\text{World Knowledge}]}\rightarrow\text{E}}{\text{worthwhile(t)} \qquad \cancel{[\neg(\text{worthwhile(t)})]^{1}}}\rightarrow\text{E}}{\perp}}{\neg\neg(\text{worthwhile(t)})}\neg\text{I}^{1}$$

$$\text{worthwhile(t)}$$

Reducing cognitive dissonance in a way that is sensitive to our self-concept, our ego, can be considered to be an instance of the psychological defense mechanism of Rationalization (Quilty-Dunn, 2022), which is possibly a component of our psychological immune system (Mandelbaum, 2019). The fact that such processes are handled at an unconscious level suggests that logic in general, and the rule of Negation Introduction in particular, are deeply embedded in our mental life.

Egan, Santos & Bloom (2007) argue that the mechanism behind cognitive dissonance may be shared with other animals. In their experiment with capuchin monkeys, subjects were first presented with pairs of food items whose baseline values had previously been established as approximately equivalent. The monkeys were then required to choose between two equally valued items, after which access to the unchosen item was blocked. They were then offered a choice between the previously unchosen item and a third item that, prior to the experiment, had been valued equally to it. The monkeys reliably avoided the previously unchosen option, suggesting that the act of choosing altered their subsequent preferences. Importantly, this effect disappeared in control conditions in which the initial exclusion of an option was imposed by the experimenter rather than resulting from the monkey's own choice. Does the change in the capuchins' attitude toward the unchosen alternative imply that they mentally employed a rule analogous to Negation Introduction? A simpler explanation may be available. Rather than involving an inferential process, the observed behavior may be the product of a mechanism of affective value distribution. On this view, given a stimulus *s*, if *s* is the object of an actionable want that remains unsatisfied, its affective value is adjusted downward and it acquires a negative valence (Russell, 2003; Barrett & Bliss-Moreau, 2009). Consequently, when *s* is later presented alongside an alternative that lacks such a valence, there may be a tendency to avoid *s* in favor of the competing option. By contrast, when access to the reward is blocked by the experimenter, the desire directed toward it is no longer actionable.[12] As a result, the reward is not experienced as the object of a frustrated desire and need not acquire a negative valence. Note that this analysis does not require the involvement of thought-related processes or any revision of the subject's beliefs, unlike the case with humans in which the negative affective valence is paired with a thought. Rather, it can be understood as a mechanism for the distribution of affective value that takes actionability into account, one that likely predates logical cognition of the kind we find in humans.

This concludes our discussion of the implementation of indirect forms of reasoning in human cognition. In the next section, we take a closer look at the mental underpinnings of the capacity to manipulate rules of inference that have access to the history of a derivation.

4. Procedural Memory and the grammar of looking back

There are important formal parallels between long-distance dependencies in syntax and suppositional states in reasoning. The representation of such processes requires access to earlier stages of a derivation. In this section, we clarify the common core of these mental operations and discuss the role that an enhanced procedural memory might have played in supporting these capacities.

Language comprises at least two subsystems: (a) the mental lexicon, a repository of basic computational units, typically morphemes, and (b) grammar, a system of rules for symbol manipulation used in the mental construction of hierarchically structured, semantically

[12] It is likely that, in such experimental settings, the experimenters are perceived by the monkeys as socially superior individuals, given their complete control over the distribution of goods. Consequently, the resources under the experimenters' control may not constitute actionable rewards in the same way as resources that are available for acquisition in natural contexts or those controlled by social subordinates.

compositional expressions. Within Ullman's declarative/procedural model (the DP model; Ullman 2004; Ullman & Pierpoint 2005), the distinction between lexicon and grammar is more than a linguistic convention: it is reflected in the contrast between declarative memory and procedural memory, two systems supported by distinct neural circuitry. According to this model, the declarative memory system underlies the mental lexicon. "This system subserves the acquisition, representation and use not only of knowledge about facts and events but also words. It stores all arbitrary, idiosyncratic word-specific knowledge, including word meanings, word sounds, and abstract representations such as word categories." (Ullman, 2004, pp. 244 – 245). The procedural memory, which is taken to be a system enabling implicit (unconscious) acquisition of motor and cognitive capacities, underlies "the computation of already-learnt, rule-based procedures that govern the regularities of language – particularly those procedures related to combining items into complex structures that have precedence (sequential) and hierarchical relations. Thus, this system is hypothesized to have an important role in rule-governed structure building" (Ullman, 2004, p. 245). Empirical evidence for the dissociation between lexicon and grammar in the brain comes from psycholinguistic (Ullman, 2001) and neuroimaging (Ullman, 2004) studies as well as from various forms of language impairment (see Ullman & Pierpoint, 2005 for an overview.)

In Minimalist syntax, Merge is the only operation responsible for building hierarchically-organized syntactic structure. This operation takes two syntactic objects, α and β, as input and defines the unique set that contains α and β as its sole members (Chomsky, 2013; Collins, 2002; Seely, 2006).

(17) Merge(α, β) = $\{\alpha, \beta\}$

When merging α and β, it may be that α is contained in β (i.e. $\beta = \{\ldots\alpha\ldots\}$, which can be read as "$\alpha$ can be found somewhere within β") or it may be that it is not contained in β. If α is not contained in β, then this subcase of Merge is called external Merge (as α is *external* to β). If α is contained in β, then this is called internal Merge (as α is *in* β).[13] Consider the derivation of the following sentences: (1) *There may be a student on the bench* and (2) *A student may be on the bench*. The relevant derivations start with merging elements of the quantificational nominal phrase and those of the prepositional phrase (Step 1, Step 2). These are all instances of external

[13] Historically, the operation of internal Merge was considered to be as an instantiation of movement transformation.

Merge. We, then, merge the nominal phrase with the preposition phrase to obtain some kind of predication structure (Step 3).

Step 1. Merge(a, student) = {a, student}

Step 2. Merge (on, Merge(the, bench)) = {on, {the, bench}}

Step 3. Merge({a, student}, {on, {the, bench}}) = {{a, student}, {on, {the, bench}}}

We add the verbal element *be* to the structure, yielding a verb phrase (Step 4). This verb phrase is then merged with the modal operator *may* (Step 5), which is interpreted as carrying present-tense information.

Step 4. Merge(be, {{a, student}, {on, {the, bench}}}) = {be, {{a, student}, {on, {the, bench}}}}

Step 5. Merge(may, {be, {{a, student}, {on, {the, bench}}}}) = {may, {be, {{a, student}, {on, {the, bench}}}}}

The English grammar has the requirement that a functional morpheme that carries the tense information (here, the tense head *may*) should be merged with a nominal element, which is called the Extended Projection Principle (EPP) property (Chomsky, 1981). We can think of this requirement as the tense head carrying a feature, the EPP feature, that must be checked via merger with a nominal phrase. There are now two ways to satisfy this requirement. We can *externally* merge 'there' to the output of Step 5 to obtain the sentence "There may be a student on the bench" (Step 6a). Alternatively, we may search into the phrase obtained in Step 5 and find a nominal phrase within it that can satisfy the merge requirement on the tense head. After scanning this syntactic representation, we detect the embedded quantificational expression *{a, student}* inside it and re-merge this nominal phrase with the structure obtained in Step 5: "$\{a, student\}_1$ may be [~~{a student}~~$_1$ on the bench]" (Step 6b). This is an instance of internal Merge. The strikethrough on the lower copy indicates that it is not to be pronounced at the level of externalization. To express the identity of the phrase *{a, student}* in both positions, we index them with the same number.

Step 6a. Merge(there, {may, {be, {{a, student}, {on, {the, bench}}}}}) (External)

Step 6b. Merge({a, student}$_1$, {may, {be, {{a, student}$_1$, {on, {the, bench}}}}}) (Internal)

Piattelli-Palmarini & Uriagereka (2005) note that a grammar that can do internal Merge is a grammar that allows access to derivation history (as in Step 6b). This capacity puts a high burden on the procedural memory (Bolender et al., 2008). In fact, the kind of procedural memory that can implement rules that make note of steps taken earlier in the derivation is likely to be uniquely human (Piattelli-Palmarini & Uriagereka, 2005, Bolender et al. 2008). In other words, a boosted procedural memory that is operative in mental manipulation of rules that can access derivational history has been claimed to be *specificially human*, as Piattelli-Palmarini & Uriagereka (2005) put it. Interestingly, the contrast between external Merge and internal Merge in terms of derivational memory is reminiscent of the contrast between direct and indirect forms of reasoning. Just like internal Merge, indirect forms of reasoning require access to earlier steps in the proof, more specifically access to the sentence under supposition. This is exemplified below in the reasoning (18), repeated from (5). While the rule of Conditional Elimination, which is an instance of a direct form of reasoning, needs no access to earlier steps in the derivation, the rule of Conditional Introduction targets the suppositional sentence introduced earlier in the proof.

(18)
$$\dfrac{\dfrac{\dfrac{\cancel{p}^{1} \qquad p \rightarrow q}{q}\,{\rightarrow}E \qquad q \rightarrow r}{r}\,{\rightarrow}E}{p \rightarrow r}\,\boxed{{\rightarrow}I^{1}}$$

If, as Piattelli-Palmarini & Uriagereka (2005) and Bolender et al. (2008) suggest, the uniqueness of internal Merge to human grammar is to be explained by the evolution of enhanced procedural memory within the human lineage, then we also get an immediate account of the uniqueness of indirect forms of reasoning to human mind. As Bolender et al. (2008, p. 143) note, the cognitive functions enabled by an enhanced procedural memory "may have played an important role in the great computational power of syntax in language," which, consequently, would explain "the qualitatively different computational, and ultimately conceptual, powers of the human mind." We shall also add that enhanced procedural memory may also underlie suppositional forms of reasoning, which are, by our hypothesis, unique to humans.

5. Conclusion

The distinction between suppositional and non-suppositional forms of reasoning is a logical one. If this paper is on the right track, it is also psychologically real. It is the distinction between the forms of reasoning that are unique to humans and those that are possibly quite widespread in the animal kingdom. This suggests, more generally, that it may be a worthwhile effort to re-analyze logical concepts in mental terms and study the consequences of such a reinterpretation. Underlying this paper is the idea that a detailed understanding of logical structures, be they classical or non-classical (Restall, 2000; Priest, 2008), should be a crucial component of the study of animal cognition in general and human cognition in particular.

In short, logic is more than a subject in discrete mathematics. It is a mind-internal capacity that we make use of in our dealings with the external world. The unique way in which we experience the world is, in part, due to the unique properties of the logic that we internalize as members of *homo sapiens*.

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